

Controlled Enablement of MCP Packages for Enterprise AI Development

1. Executive Summary

AI adoption across the enterprise is evolving rapidly, with increasing interest in MCP-based capabilities for Agentic and GenAI solutions. Current controls are intentionally restrictive due to evolving technology and limited monitoring for MCP usage. This document proposes a controlled enablement model using Active Directory (AD) groups, enabling identity-based access, governance, and auditability.

2. Current State

Artifact Source: Nexus Repository (validated/scanned packages)
Consumption: End-user devices and GitLab CI/CD pipelines
Control: Pattern-based restriction on *mcp* packages
Observation: Growing demand for MCP-based libraries and structured access requirements

3. Problem Statement

Current controls provide broad coverage but lack granularity. They do not distinguish between trusted vs exploratory usage, end-user vs pipeline execution, or approved vs unapproved workloads. A more precise identity-driven model is required.

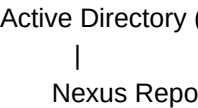
4. Risk Analysis

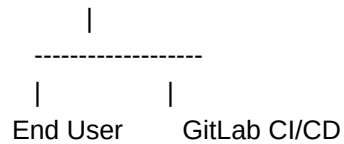
Risk Area	Description
Supply Chain Risk	Unverified MCP packages
Data Exposure	External interactions
Execution Risk	Dynamic code execution
Observability Gap	Limited visibility
Access Misalignment	Over-permissioned access

5. Proposed Solution

Use AD groups for controlled access:
End Users: SG-GOZ-<AssetID>-AIExploration-MCP-Users
Pipelines: SG-GOZ-<AssetID>-AIExploration-MCP-Pipeline
Nexus validates membership before allowing access.

6. Architecture Overview





7. Security Controls

Preventive: AD-based restriction, scanning

Detective: Logging and SIEM integration

Corrective: Access removal and package restriction

8. Governance Model

Onboarding approval, periodic reviews, centralized monitoring, time-bound exceptions

9. Benefits

Enables controlled AI adoption, improves governance, ensures auditability, aligns with security principles

10. Future Enhancements

Runtime monitoring, egress controls, behavioral analytics, policy-as-code

11. Decision Required

Approve AD-based controlled access model for MCP packages for users and pipelines.

12. Bottom Line

Identity-driven access with governance enables secure and scalable MCP adoption.